

Ching-Wen Hung

Ph.D. Candidate in Computer Science
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Research Interests

- Human-Computer Interaction (HCI), Embodied Interaction, Haptic and Gesture Interfaces, Virtual/Augmented Reality (VR/AR), Perception and Inclusive Interactive Systems

Education

- **National Taiwan University** **Taipei, Taiwan**
Ph.D. in Computer Science (Direct Ph.D. Program) *Mar. 2020 - Present*
Advisor: Bing-Yu Chen
- **National Taiwan University** **Taipei, Taiwan**
Graduate Coursework in Information Management *Sep. 2018 - Feb. 2020*
Entered Direct Ph.D. Program in Computer Science
Advisor: Bing-Yu Chen
- **National Central University** **Taoyuan, Taiwan**
Bachelor of Business Administration in Information Management *Sep. 2014 - Jun. 2018*

Research Experiences

- **Visiting Faculty Assistant - Embodied Dynamics Laboratory** *Oct. 2023 - Sep. 2024*
University of Maryland, College Park, Maryland, USA
Supervisor: Jun Nishida
- **Research Assistant - Communications and Multimedia Laboratory** *Sep. 2018 - Present*
National Taiwan University, Taipei, Taiwan
Supervisor: Bing-Yu Chen

Publications

- **DilatArm: An Accessible, Low-cost Passive Exoskeleton using Dilatant Fluid for Viscosity Feedback during Spatial Movement**
Ching-Wen Hung, Eric Chen, Hamza Shah Khan, Po-Ying Hsu, Bing-Yu Chen, Jun Nishida
In Proceedings of the ACM Symposium on Spatial User Interaction (SUI 2026), to appear. (oral present)
- **ElderPlay: Supporting Age-Inclusive Gameplay for Older Adults via Real-Time Gesture-to-Controller Translation**
Ching-Wen Hung, Che-Wei Hsu, Wei-Tang Hsu, Tzu-Chin Chiu, Li Lin, Yao Cheng Lee, Ting-Wu Chang, Hsien-Hui Tang, Bing-Yu Chen, Mike Y. Chen
In Proceedings of the ACM Designing Interactive Systems Conference (DIS 2026) (oral present) [accepted rate: 21%]
- **TacNote: Tactile and Audio Note-Taking for Non-Visual Access**
Wan-Chen Lee, **Ching-Wen Hung**, Chao-Hsien Ting, Peggy Chi, Bing-Yu Chen
In Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2023). (oral present) [accepted rate: 25.1%]
- **Puppeteer: Exploring Intuitive Hand Gestures and Upper-Body Postures for Manipulating Human Avatar Actions**
Ching-Wen Hung, Ruei-Che Chang, Hong-Sheng Chen, Chung-Han Liang, Liwei Chan, Bing-Yu Chen
In Proceedings of the ACM Symposium on Virtual Reality Software and Technology (VRST 2022). (oral present) [accepted rate: 26.7%]

- **OsciHead: Simulating Versatile Force Feedback on an HMD by Rendering Various Types of Oscillation**
Ching-Wen Hung, Hsin-Ruey Tsai, Chi-Chun Su, Jui-Cheng Chiu, Bing-Yu Chen
In Proceedings of the ACM Human-Computer Interaction with Mobile Devices and Services (MobileHCI 2022).
In Proceedings of the ACM on Human-Computer Interaction (PACMHCI 2022). (oral present) [accepted rate: 36%]
- **ElastOscillation: 3D Multilevel Force Feedback for Damped Oscillation on VR Controllers**
Hsin-Ruey Tsai, **Ching-Wen Hung**, Tzu-Chun Wu, Bing-Yu Chen
In Proceedings of the ACM Human-Computer Interaction (CHI 2020). (oral present) [accepted rate: 24.31%]

Projects, Posters, Demos

- **Designing Embodied Interaction: Shaping Bodily Action, Experience, and Capability**
Ching-Wen Hung
In Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2026 Doctoral Symposium).
- **Demonstration of DilatArm: An Accessible, Low-Cost Passive Exoskeleton Powered by Dilatant Fluid for Velocity-Dependent Force Feedback**
Eric Chen, **Ching-Wen Hung**, Hamza Shah Khan, Po-Ying Hsu, Bing-Yu Chen, Jun Nishida
In Proceedings of the ACM Symposium on Spatial User Interaction (SUI 2026 Demo).
- **FingerPuppet: Finger-Walking Performance-based Puppetry for Human Avatar**
Ching-Wen Hung, Chung-Han Liang, Bing-Yu Chen
In Proceedings of the ACM Human-Computer Interaction (CHI 2024 Late-Breaking Work). [accepted rate: 33.88%]
- **Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper-Body Postures**
Ching-Wen Hung, Ruei-Che Chang, Hong-Sheng Chen, Chung Han Liang, Liwei Chan, Bing-Yu Chen
In Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2022 Adjunct).
- **Tonopoly: A Monopoly-like Board Game with Toio**
Chi-Huan Chiang, **Ching-Wen Hung**
In Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2021 Student Innovation Contest).
- **Demonstration of ElastOscillation: A VR Controller Providing 3D Multilevel Feedback for Damped Oscillation**
Ching-Wen Hung, Hsin-Ruey Tsai, Tzu-Chun Wu, Bing-Yu Chen
In Proceedings of the ACM Human-Computer Interaction (CHI 2020 Extended Abstracts).

Invited Talks and Presentations

- **Embodied Interaction: Enhancing Immersive Interaction in Virtual and Physical Realities**
Seminar Talk at National Chung Cheng University (Chiayi, Taiwan) 2024
- **PhD Research in HCI: Exploring Embodied Interaction and Funding Opportunities in Taiwan**
Podcast (Online) 2024
- **TacNote: Tactile and Audio Note-Taking for Non-Visual Access**
UIST 2023 (San Francisco, California, USA)
TAICHI 2023 (Taipei, Taiwan)
- **Puppeteer: Exploring Intuitive Hand Gestures and Upper-Body Postures for Manipulating Human Avatar Actions**
VRST 2022 (Tsukuba, Japan)
TAICHI 2022 (Hsinchu, Taiwan)
- **Puppeteer: Manipulating Human Avatar Actions with Intuitive Hand Gestures and Upper-Body Postures**
UIST 2022 Adjunct (Bend, Oregon, USA)

- **OsciHead: Simulating Versatile Force Feedback on an HMD by Rendering Various Types of Oscillation**
MobileHCI 2022 (Online Virtual Conference)
TAICHI 2021 (Online Virtual Conference)
- **ElastOscillation: 3D Multilevel Force Feedback for Damped Oscillation on VR Controllers**
CHI 2020 (Online Virtual Conference)
UTokyo-NTU Joint Conf 2019 (Tokyo, Japan)

Teaching Certifications

- English-Medium Instruction (EMI) Teaching Assistant Certification, National Taiwan University, 2026
- Mandarin-Medium Teaching Assistant (TA) Certification, National Taiwan University, 2020

Teaching Experiences

• Teaching Assistant

- **Multimodal Human-Computer Interaction** *National Taiwan University* *Feb. 2026 - Jun. 2026*
Instructor: Shan-Yuan Teng
 - * English-medium instruction (EMI) course
 - * Led hands-on workshops on haptic interaction and robotic arm prototyping using MeArm kits, including servo calibration, Arduino control, and system assembly
 - * Designed and 3D-printed customized components to support student assignments and facilitate exploration of haptic applications
 - * Mentored students in multimodal prototyping across visual, audio, and haptic modalities, providing technical guidance on Processing, Arduino, and hardware integration
 - * Advised final course projects on interaction design, research methodology, and user study planning, including proposal feedback and experimental design review
 - * Evaluated student mini-project expos and final projects by providing design critique, usability feedback, and research-oriented assessment across multimodal interaction assignments
- **Game Programming** *National Taiwan University* *Sep. 2025 - Dec. 2025*
Instructor: Ken-Yi Lee, Bing-Yu Chen
 - * Mandarin-medium instruction course
 - * Assisted with grading and course administration
 - * Coordinated the final project demo workshop, facilitating student presentations and assessment by instructors and invited industry professionals
- **Human Computer Interaction in Industry** *National Taiwan University* *Sep. 2022 - Dec. 2022*
Instructor: Peggy Chi
 - * English-medium instruction (EMI) course
 - * Facilitated student discussions and project development for industry-oriented HCI topics
 - * Provided feedback and guidance on final project presentations
- **Creative and Design Thinking** *National Taiwan University* *Mar. 2020 - Jun. 2020*
Instructor: Bing-Yu Chen
 - * Mandarin-medium instruction course for executive entrepreneurship program students (NTU EiMBA)
 - * Facilitated two-day design thinking workshops
 - * Assisted with team formation, activity coordination, and final presentations
- **Internet Technology in E-Commerce** *National Central University* *Sep. 2017 - Jan. 2018*
Instructor: Kuen-Liang Su, Shi-Jen Lin
 - * Mandarin-medium instruction course (In-service Master's Program)
 - * Supported classroom coordination and course administration for working professionals
 - * Assisted with group organization, attendance tracking, and grading records

Student Mentoring

- **Bo-Ru Chen***, *Master Student*, National Taiwan University *Jul. 2025 - Present*
- **Che-Wei Hsu***, *Undergraduate Student*, National Taiwan University *Jul. 2025 - Jun. 2026*
- **Wei-Tang Hsu***, *Undergraduate Student*, National Taiwan University *Jul. 2025 - Jun. 2026*
- **Po-Ying Hsu***, *Master Student*, National Taiwan University *Dec. 2024 - Jul. 2025*
- **Wan-Chen Lee***, *Master Student*, National Taiwan University *Dec. 2022 - Oct. 2023*
- **Min-Rui Li**, *Master Student*, National Taiwan University *Feb. 2023 - Jun. 2023*
- **Wei-Xun Huang**, *Master Student*, National Taiwan University *Feb. 2023 - Jun. 2023*
- **Hong-Sheng Chen***, *Master Student*, National Taiwan University *Nov. 2021 - Apr. 2022*
- **Chung Han Liang***, *Master Student*, National Taiwan University *Nov. 2021 - Apr. 2022*
- **Chi-Chun Su***, *Master Student*, National Taiwan University *Aug. 2020 - Aug. 2021*
- **Jui-Cheng Chiu***, *Undergraduate Student*, National Taiwan University *Aug. 2020 - Apr. 2021*

* Co-authored peer-reviewed publications.

Academic and Co-Curricular Teaching

- Programming Instructor and Executive Member, NCU Big Data and Programming Society *2016–2017*
 - Delivered programming workshops (Web, Android, Linux) for undergraduate students
 - Developed instructional materials and hands-on coding exercises
- C Programming Instructor, 21st NCU MIS Camp *2015*
 - Designed and delivered introductory C programming workshops for high school students
 - Developed course materials and interactive programming activities

Awards and Grants

- **Ph.D. Students Study Abroad Program** *2023 - 2024*
Supported one-year visiting research appointment at the Department of Computer Science, University of Maryland, College Park
(approximately USD 30,000 funding support)
National Science and Technology Council, Taiwan
- **UIST Travel Award** *2022*
Received travel grant support of USD 4,433 for presenting research at ACM UIST 2022
ACM UIST
- **Gary Marsden Travel Award** *2022*
(Selected recipient; declined in favor of ACM UIST Travel Grant)
ACM SIGCHI
- **Subsidy for Domestic Graduate Students to Attend International Academic Conferences** *2022*
National Science and Technology Council, Taiwan
- **Fellowship for Outstanding Ph.D. Student** *2020 - 2024*
Awarded monthly fellowship support for outstanding Ph.D. students
Ministry of Science and Technology, Taiwan
- **Fellowship for Students of the Direct Pursuit of Ph.D. Degree** *2020 - 2020*
Awarded monthly fellowship support upon entering the Direct Ph.D. Program
National Taiwan University, Taiwan
- **Dean's List Award** *2015 - 2017*
(Top academic performance, 2015–2017)
National Central University, Taiwan

Leadership

- Coordinator, Taiwan–Japan HCI Joint Workshop 2026
- Session Chair, Taiwan HCI Internal Pre-CHI Workshop 2026
- Project Leader, CMLab (Robin Group) 2020 - 2023, 2026
- Instructor, CMLab (Robin Group) HCIxGraphics Workshop 2020 - 2022
- Instructor, CMLab (Robin Group) HCI Workshop 2019
- Host, TAICHI 2022 – Women in STEM Luncheon 2022
- Host, TAICHI x UIST Online Meeting 2021

Professional Services

- **Conference Committee Roles**
 - TAICHI 2026 – Demo Chair
 - TAICHI 2025 – Demo Chair
 - TAICHI 2023 – Demo Chair
 - TAICHI 2022 – Program Chair
- **Reviewer**
 - VRST 2026
 - TAICHI 2026 (Demo)
 - AHs 2026
 - CHI 2026 (Papers & Posters)
 - SIGGRAPH Asia 2025 (ET)
 - TAICHI 2025 (Demo)
 - CGW 2025
 - UIST 2025
 - CHI 2025
 - TAICHI 2024
 - CHI 2024 (LBW)
 - TAICHI 2023
 - CHI 2023 (LBW)
 - AHs 2023
 - TAICHI 2022
 - TAICHI 2021
- **Student Volunteer**
 - Ubicomp 2021
 - CHI 2021
 - TAICHI 2019

References

- **Bing-Yu Chen**
Distinguished Professor at Department of Computer Science and Information Engineering / Department of Information Management / Graduate Institute of Networking and Multimedia, National Taiwan University
Email: robin@ntu.edu.tw
- **Jun Nishida**
Assistant Professor at Department of Computer Science, University of Maryland, College Park
Email: jun@umd.edu
- **Shan-Yuan Teng**
Assistant Professor at Department of Computer Science & Information Engineering, National Taiwan University
Email: tengshanyuan@csie.ntu.edu.tw

- **Mike Y. Chen**
Professor of Computer Science, National Taiwan University
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- **Hsien-Hui Tang**
Professor of Department of Commercial and Industrial Design, National University of Science and Technology
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- **Liwei Chan**
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